

Experiment:18

Construct a C program to simulate producer consumer problem using semaphores.

Code:-

```
#include <stdio.h>

#include <pthread.h>

#include <semaphore.h>

#ifdef _WIN32

#include <windows.h>

#define SLEEP(ms) Sleep(ms)

#else #include <unistd.h>

#define SLEEP(ms) usleep((ms) * 1000)

#endif

#define BUFFER_SIZE 5 #define MAX_ITEMS 10

int buffer[BUFFER_SIZE] = {0};

sem_t empty, full;

int produced_items = 0; int consumed_items = 0;

void *producer(void *arg) { while (produced_items < MAX_ITEMS)

{ sem_wait(&empty);

for (int i = 0; i < BUFFER_SIZE; i++)
{
if (buffer[i] == 0)
{
buffer[i] = produced_items + 1;
printf("Produced: %d\n", buffer[i]);
produced_items++;
break;
}
}
```

```

    }

    sem_post(&full);
    SLEEP(1000);
}

return NULL;

}

void *consumer(void *arg) { while (consumed_items < MAX_ITEMS) { sem_wait(&full);

    for (int i = 0; i < BUFFER_SIZE; i++)
    {
        if (buffer[i] != 0)
        {
            printf("Consumed: %d\n", buffer[i]);
            buffer[i] = 0;
            consumed_items++;
            break;
        }
    }

    sem_post(&empty);
    SLEEP(2000);
}

return NULL;

}

int main() { pthread_t producer_thread, consumer_thread;

sem_init(&empty, 0, BUFFER_SIZE);
sem_init(&full, 0, 0);

pthread_create(&producer_thread, NULL, producer, NULL);
pthread_create(&consumer_thread, NULL, consumer, NULL);

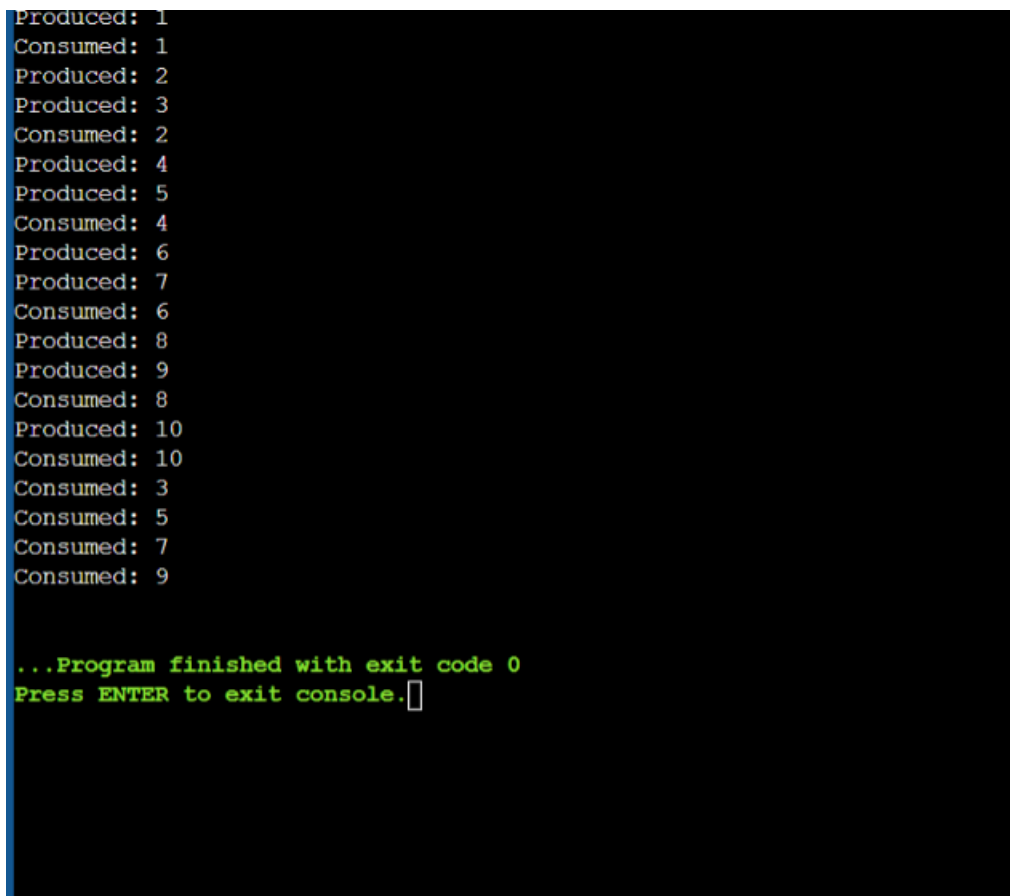
```

```
pthread_join(producer_thread, NULL);  
pthread_join(consumer_thread, NULL);
```

```
sem_destroy(&empty);  
sem_destroy(&full);
```

```
return 0;  
}
```

Output:-



```
Produced: 1  
Consumed: 1  
Produced: 2  
Produced: 3  
Consumed: 2  
Produced: 4  
Produced: 5  
Consumed: 4  
Produced: 6  
Produced: 7  
Consumed: 6  
Produced: 8  
Produced: 9  
Consumed: 8  
Produced: 10  
Consumed: 10  
Consumed: 3  
Consumed: 5  
Consumed: 7  
Consumed: 9  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```